CHE 201 Fall 2019 Final Exam

Angela Tangi Ester Sianturi

TOTAL POINTS

76 / 100

QUESTION 1

1 15 pts

1.1 a **8 / 12**

- **- 0 pts** Correct
- **- 2 pts** first answer is incorrect
- **- 2 pts** second answer incorrect
- **- 2 pts** third answer is incorrect
- ✓ **- 2 pts fourth answer is incorrect**
- ✓ **- 2 pts fifth answer is incorrect**
- **- 2 pts** sixth answer is incorrect

1.2 b **0 / 3**

- **- 0 pts** Correct
- ✓ **- 3 pts incorrect**
- **- 2 pts** explanation is wrong

QUESTION 2

2 10 pts

2.1 a **5 / 5**

- ✓ **- 0 pts Correct**
- **- 2 pts** formula is wrong
- **- 5 pts** incorrect
- **- 2 pts** temperature is not converted to K

2.2 b **5 / 5**

- ✓ **- 0 pts Correct**
- **- 5 pts** incorrect

QUESTION 3

10 pts

3.1 a **0 / 2**

- **- 0 pts** Correct
- ✓ **- 2 pts incorrect**
- **- 1 pts** negative sign missing

3.2 b **2 / 2**

- ✓ **- 0 pts Correct**
- **- 2 pts** incorrect

3.3 c **6 / 6**

- ✓ **- 0 pts Correct**
- **- 1 pts** calculation error
- **- 5 pts** Incorrect or no values
- **- 6 pts** Incorrect or no answer
- **- 2 pts** Did not multiply by mass
- **- 2 pts** Wrong value for u
- **- 3 pts** Did not show the calculations for the interpolate values
- **- 2 pts** T should be in Kelvin

QUESTION 4

32 pts

4.1 a **2 / 2**

- ✓ **- 0 pts Correct**
- **- 1 pts** boundary is wrong

4.2 b **3 / 3**

- ✓ **- 0 pts Correct**
- **- 1 pts** closed system missing
- **- 1 pts** constant pressure missing
- **- 1 pts** $\Delta KE = \Delta PE = 0$ missing

4.3 c **3 / 4**

- ✓ **- 0 pts Correct**
- **- 2 pts** Wrong simplified energy balance
- **- 2 pts** Wrong simplified entropy balance
- **- 1 pts** Need to simplify the entropy balance

- 1 Point adjustment

delta S is not zero

4.4 d **8 / 8**

✓ **- 0 pts** Correct

- **1 pts** unit conversion for work
- **2 pts** v2 is wrong
- **1 pts** P is wrong
- **1 pts** Work is not multiplied with mass
- **4 pts** Calculation of Q is wrong
- **4 pts** Calculation of W is wrong
- **8 pts** Wrong answer
- **2 pts** u2 is wrong
- **1 pts** U1 is wrong
- **1 pts** v1 is wrong
- **1 pts** Calculation error
- **2 pts** States 1 and 2 are inverted

4.5 e **5 / 5**

✓ **- 0 pts** Correct

- **1 pts** need to multiply by mass
- **5 pts** Wrong or no answer
- **1 pts** Should use Table A-2 or interpolate Table A-3
- **2 pts** s2 is wrong
- **2 pts** s1 is wrong
- **1 pts** Calculation error

4.6 f **4 / 5**

- **0 pts** Correct
- **1 pts** T is wrong should be boundary T
- **2 pts** Did not state if process is reversible, irreversible or impossible
- **3 pts** Wrong solution

✓ **- 1 pts Process is impossible**

- **1 pts** Calculation or equation error
- **5 pts** No answer

4.7 g **5 / 5**

✓ **- 0 pts** Correct

- **1 pts** x axis not labeled
- **2 pts** T-s graph is wrong or absent
- **1 pts** v values are wrong or absent
- **1 pts** s value is wrong or absent
- **2 pts** P-v graphs is wrong
- **1 pts** y axis not labeled
- **1 pts** Wrong or missing units

QUESTION 5

5 33 pts

5.1 a **4 / 4**

✓ **- 0 pts** Correct

- **4 pts** Incorrect
- **1 pts** Did not represent Work
- **1 pts** Wrong system boundary
- **1 pts** Missing one outlet

5.2 b **3 / 5**

- **0 pts** Correct
- **1 pts** steady state missing

✓ **- 2 pts control volume missing**

- **1 pts** $\Delta PE = 0$ missing
- **1 pts** Cannot assume $\Delta KE = 0$

5.3 c **4 / 6**

- **0 pts** Correct

✓ **- 2 pts entropy balance is wrong**

- **2 pts** energy balance is wrong
- **2 pts** Q, W, m is not written in rate format
- **1 pts** Need to simplify entropy balance

5.4 d **9 / 10**

- **0 pts** Correct

✓ **- 1 pts calculation eror**

- **1.5 pts** unit conversion error
- **2 pts** h1 is wrong
- **2 pts** h2 is wrong
- **2 pts** h3 is wrong
- **8 pts** Incomplete answer
- **1 pts** Answer should be in MW
- **1 pts** m1 is wrong
- **2 pts** m2 and m3 are wrong
- **1 pts** incomplete calculation

5.5 e 0 / 8

- **- 0 pts** Correct
- ✓ **- 8 pts** **incorrect or no answer**
- **- 1 pts** calculation error
- **- 2 pts** s2 is wrong or absent
- **- 2 pts** s3 is wrong or absent
- **- 2 pts** T is wrong
- **- 4 pts** Incorrect equation or no equation
- **- 2 pts** m2 and m3 are wrong or absent
- **- 1 pts** unit conversion error
- **- 2 pts** s1 is wrong

CHE 201 Final Exam **NAME:** Angela Sianturi

Bilgin December 9th, 2019

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (15 pts)

True or False (2 pts each)

___T___The upstream and downstream enthalpies of flow through an insulated valve is equal.

___F___Entropy, S, is path-dependent. independent

___F___Gibbs energy is an intensive property. extencive

___T___A closed system can experience an increase in entropy only when there is heat transfer from the system. $\Delta S = \int \frac{Q}{T} + \sigma$ → heat transfer.

___F___The change in Gibbs free energy (dg) is equal to zero for a phase change from a saturated liquid to a saturated vapor.

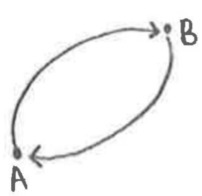
___T___The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

Fill in the blank

(3 pts) A gas goes from State A to State B in a reversible, adiabatic process. It then goes from State B back to State A by a different pathway that is irreversible and not adiabatic. The entropy change, $\boxed{(\Delta S)}$ for the irreversible pathway is

___less than___ (greater than/equal to/ less than) zero. Explain.

since A → B is adiabatic then $Q_{A \to B} = 0$



$$\Delta S = \int_1^2 \frac{Q}{T} + \sigma$$

$$0 = \frac{Q_2}{T_2} - \cancel{\frac{Q_1}{T_1}}^{\,0} + \sigma$$

$$0 = \frac{Q_2}{T_2} + \sigma$$

$$-\sigma = \frac{Q_2}{T_2}$$

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QUESTION 2 (10 pts)

a) (5 pts) A steam engine operates with a theoretical maximum efficiency of %50 and a minimum temperature of 150°C. Calculate temperature of the engine to maintain maximum efficiency.

$$COP = 50\% = \frac{5\cancel{0}}{10\cancel{0}} = 0.5$$

$$0.5 = \frac{Th - Tc}{Th}$$

$$0.5\,Th = Th - (150 + 273)$$

$$423 = 0.5\,Th$$

$$\boxed{Th = 846\text{ K}} \text{ or } \boxed{Th = 573°C}$$

b) (5 pts) The Reynolds number, Re, is a dimensionless number that measures the ratio of inertial forces to viscous forces in fluid flow;

$$Re = \frac{\rho V L}{\mu}$$

In this equation, ρ is the density (kg/m³), V is the velocity (m/s), L is a characteristic length (m), and μ is the viscosity. What are the dimensions of μ? **Show all your work for full credit.**

dimensionless

$$Re = \frac{\rho V L}{\mu} \qquad \mu = \rho \cdot V \cdot L$$

$$\mu = \frac{kg}{m^{\cancel{3}}} \cdot \frac{\cancel{m}}{s} \cdot \cancel{m}$$

$$\boxed{\mu = \frac{kg}{m \cdot s}}$$

QUESTION 3 (10 pts)

a) (2 pts) Write the specific Maxwell relation that corresponds to the thermodynamic function:

$$du = \underset{M}{T}\,\underset{dx}{ds} - \underset{N}{p}\,\underset{dy}{dv} \qquad \left(\frac{dM}{dy}\right)_x = \left(\frac{dN}{dx}\right)_y$$

u (s, v): $\left(\frac{\partial T}{\partial p}\right)_s = \left(-\frac{\partial P}{\partial s}\right)_v$

b) (2 pts) Write the two, second mixed partial derivatives for the exact differential:

$$\frac{\partial h^2}{\partial s \partial p} = \left(\frac{\partial T}{\partial p}\right)_s \qquad \frac{\partial h^2}{\partial p \partial s} = \left(\frac{\partial v}{\partial s}\right)_p$$

dh = $\underset{M}{T}\,\underset{dx}{ds} + \underset{N}{v}\,\underset{dy}{dP}$: $\left(\frac{\partial T}{\partial p}\right)_s = \left(\frac{\partial v}{\partial s}\right)_p$

b) (6 pts) A 3 kg mass of saturated liquid water has a pressure of 10.0 bar. What is the Helmholtz energy, in kJ?

Show all your work for full credit

$$\boxed{\psi = u - Ts}$$

@ 10 bar , sat liquid $\rightarrow u_f = 761.68\ kJ/kg$

$T = 179.9°C = 453.05\ K$

$s_f = 2.1387\ \frac{kJ}{kg \cdot K}$

$$\psi = \left[761.68\ \frac{kJ}{\cancel{kg}} - \left(453.05\ \cancel{K} \cdot 2.1387\ \frac{kJ}{\cancel{kg} \cdot \cancel{K}}\right)\right] \cdot 3\ \cancel{kg}$$

$$\boxed{\psi = -621.77\ kJ}$$

QUESTION 4: (32 pts)

3 kg of water, initially a saturated liquid at 120°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 25°C room until the water becomes a state where x=0.8. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

a) (2 pts) Draw a schematic of the system and label the system boundary.

b) (3 pts) State the engineering model

c) (4 pts) Write the simplified energy balance and simplified entropy balance.

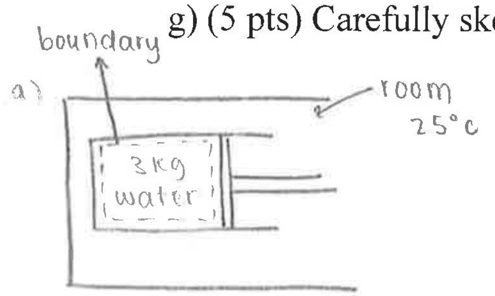
d) (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.

e) (5 pts) Determine the change in entropy of the water during the process, **ΔS, in kJ/K**

f) (5 pts) Determine the entropy generated during the process, **σ, in kJ/K.** State if this process is reversible, irreversible or impossible.

g) (5 pts) Carefully sketch the process on P-v and T-s diagram.

a)



b) engineering model:

- piston - cylinder
- in a room
- piston moves freely ($W \neq 0$)
- constant pressure
- $\Delta KE = \Delta PE = 0$
- heat transfer
- closed system

c) $Q - W = \cancel{\Delta PE}^{0} + \cancel{\Delta KE}^{0} + \Delta U$

$\boxed{Q - W = \Delta U}$ → energy balance

$\Delta S = \int_1^2 \frac{\delta Q}{T} + \sigma$

$\boxed{0 = \int_1^2 \frac{\delta Q}{T} + \sigma}$ → entropy balance

$= 1.155 \times 10^{-7}$

d) $W = \int_{V_1}^{V_2} p\,dV = p \int_{V_1}^{V_2} dV = p(V_2 - V_1)$

@ state 1 → 120°C, sat liquid

$p = 1.985$ bar

$v_1 = 1.0603 \times 10^{-3} \frac{m^3}{kg}$

$V_1 = 1.0603 \times 10^{-3} \frac{m^3}{\cancel{kg}} \times 3\cancel{kg}$

$= 3.1809 \times 10^{-3}\ m^3$

@ state 2 → $x = 0.8$, $p = 1.985$ bar ≈ 2 bar

$v_f = 1.0605 \times 10^{-3} \frac{m^3}{kg}$

$v_g = 0.8857 \frac{m^3}{kg}$

$v_2 = v_f + x(v_g - v_f)$

$= (1.0605 \times 10^{-3}) + (0.8)(0.8857 - 1.0605 \times 10^{-3})$

$= 0.708 \frac{m^3}{kg}$

$V_2 = 0.708 \frac{m^3}{kg} \times 3\ kg = 2.1263\ m^3$

k hd md

$W = P(V_2 - V_1)$

$= (1.985\ \cancel{bar})\left(\frac{10^2\ \cancel{kPa}}{1\ \cancel{bar}}\right)\left(\frac{1000\ \cancel{Pa}}{1\ \cancel{kPa}}\right)(2.1263 - 3.1809\times10^{-3})m^3\left(\frac{1\ N}{\cancel{m^2}\cdot 1\ \cancel{Pa}}\right)$

$= 421,439\ J\left(\frac{1\ kJ}{1000 J}\right)$

$\boxed{W = 421.44\ kJ}$

@ state 1 $\rightarrow U_f = 503.50\ \frac{kJ}{\cancel{kg}} \times 3\ \cancel{kg} = 1510.5\ kJ$

@ state 2 $\rightarrow U_f = 504.49\ \frac{kJ}{kg}$

$U_g = 2529.5\ \frac{kJ}{kg}$ $\longrightarrow U_2 = U_f + x(U_g - U_f)$

$= 504.49 + (0.8)(2529.5 - 504.49)$

$= 2124.5\ \frac{kJ}{\cancel{kg}} \times 3\ \cancel{kg} = 6373.494\ kJ$

$Q = \Delta U + W$

$= (U_2 - U_1) + W \Rightarrow (6373.494 - 1510.5) + 421.44\ kJ$

$\boxed{Q = 5284.434\ kJ}$

e) @ state 1 $\rightarrow S_f = 1.5276\ \frac{kJ}{\cancel{kg}\cdot k} \times 3\ \cancel{kg} = 4.5828\ \frac{kJ}{K}$

@ state 2 $\rightarrow S_f = 1.5301\ \frac{kJ}{kg\cdot k}$ $\quad S_2 = 1.5301 + (0.8)(7.1271 - 1.5301)$

$S_g = 7.1271\ \frac{kJ}{kg\cdot K}$ $\quad = 6.0077\ \frac{kJ}{\cancel{kg}\cdot K} \times 3\ \cancel{kg} = 18.0231\ \frac{kJ}{k}$

$\Delta S = S_2 - S_1 = 18.0231 - 4.5828$

$\boxed{\Delta S = 13.4406\ \frac{kJ}{k}}$

f) $\Delta S = \int \frac{\delta Q}{T} + \sigma$

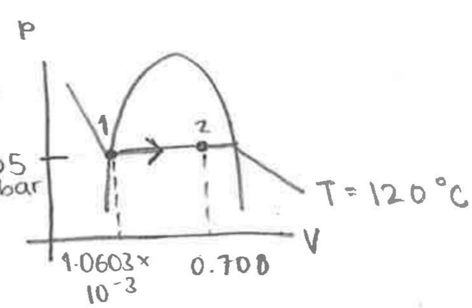
$13.4406\ \frac{kJ}{k} = \frac{5284.434\ kJ}{(273.15 + 25)k} + \sigma$

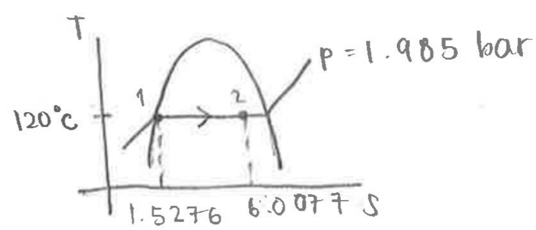
$13.4406\ \frac{kJ}{k} - 17.72407\ \frac{kJ}{k} = \sigma$

$\boxed{-4.2834\ \frac{kJ}{k} = \sigma}$

$\sigma < 0$, the process is reversible

g)





$0.1\,MW = 10^2\,kW$

$1\,MW = 1000\,kW$ $\qquad W = \frac{J}{S}$

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Nm

QUESTION 5: (33 pts)

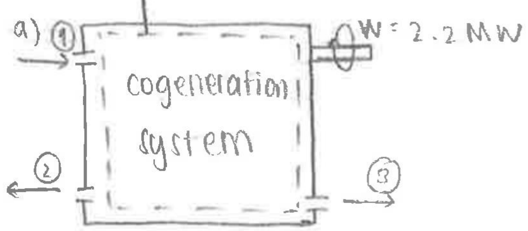
→open system

A cogeneration system operates at steady state. Water vapor enters the system at 20 bar, 360°C, velocity of 50m/s and mass flow rate of 1.5kg/s. Power is developed by the system at a rate of 2.2MW. There are two outlets for the system: process steam and hot water. Process steam leaves as saturated vapor at 1 bar, 100m/s, and hot water leaves as saturated liquid at 1 bar, 100m/s. Mass flow rates of process steam and hot water are equal. Neglect any changes in potential energy.

a) (4pts) Draw a schematic of the system and label the system boundary.
b) (5pts) State the engineering model.
c) (6pts) Write simplified balances for mass, energy and entropy.
d) (10pts) Calculate the rate of heat transfer, in MW, between the system and its surroundings.
e) (8pts) Calculate the entropy production for the process, in kJ/K-s.

system boundary

a)



b) engineering model:
- cogeneration system
- one inlet, two outlet
- steady state
- power developed ($W \neq 0$)
- $\Delta PE = 0$

① Water vapor
$P = 20$ bar $\quad V_1 = 50\,m/s$
$T = 360°C \quad \dot m_1 = 1.5\,kg/s$
$h_1 = 3159.3\,kJ/kg$

② saturated vapor
$P = 1$ bar
$V_2 = 100\,m/s$
$h_2 = 2675.5\,kJ/kg$

③ saturated liquid
$P = 1$ bar
$V_3 = 100\,m/s$
$h_3 = 417.46\,kJ/kg$

c) mass balance:

$$\cancel{\frac{dm_{cv}}{dt}}^{0} = \dot m_i - \dot m_e \qquad \dot m_2 = \dot m_3$$

$$\boxed{0 = \dot m_1 - (\dot m_2 + \dot m_3)}$$

$\dot m_1 = 2\dot m_e$

energy balance:

$$\cancel{\frac{dE_{cv}}{dt}}^{0} = \dot Q_{cv} - \dot W_{cv} + \sum \dot m_i\left(h_i + \frac{V_i^2}{2} + \cancel{gz_i}^{0}\right) - \sum \dot m_e\left(h_e + \frac{V_e^2}{2} + \cancel{gz_e}^{0}\right)$$

$$\boxed{0 = \dot Q_{cv} - \dot W_{cv} + \dot m_1\left(h_1 + \frac{V_1^2}{2}\right) - \dot m_2\left(h_2 + \frac{V_2^2}{2}\right) - \dot m_3\left(h_3 + \frac{V_3^2}{2}\right)}$$

d) $\dot m_1 = 2\dot m_e$

$1.5\,\frac{kg}{s} = 2\dot m_e$

$\frac{1.5}{2} = \dot m_e = 0.75\,\frac{kg}{s}$

$$0 = \dot Q_{cv} - 2.2\,MW + \left(1.5\,\frac{kg}{s}\right)\left(3159.3\,\frac{kJ}{kg} + \frac{(50)^2}{2} \times \frac{1}{1000}\right) - \left(0.75\,\frac{kg}{s}\right)\left(2675.5\,\frac{kJ}{kg} + \frac{(100)^2}{2} \times \frac{1}{1000}\right) - \left(0.75\,\frac{kg}{s}\right)\left(417.46\,\frac{kJ}{kg} + \frac{(100)^2}{2} \times \frac{1}{1000}\right)$$

$-\dot Q_{cv} = -2.2\,MW + 4740.825\,MW - 2010.375\,MW - 316.845\,MW$

$-\dot Q_{cv} = 2411.405\,MW$

$$\boxed{\dot Q_{cv} = -2411.405\,MW}$$

e) $\Delta S = \int \frac{\delta Q}{T} + \sigma$

$-2411.405 \cancel{MW} \left(\frac{1000 \text{ kW}}{1 \cancel{MW}} \right)$

$= -2411405 \text{ kW}$

$0 = \frac{Q}{T} + \sigma$

$-\sigma = \frac{Q}{T}$

$\cancel{-}\sigma = \frac{\cancel{-}2411405 \text{ kJ}}{s \cdot (360 + 273.15) K}$

$$\boxed{\sigma = 3808.58 \frac{kJ}{K \cdot s}}$$